HW3

2025-02-21

#Problem 1

```r
library(readxl)
library(boot)
```

```r
birth_dat <- read_excel("/Users/manideeppendyala/Downloads/birth_dat (1).xlsx")
print(birth_dat,n = 10)
```

```
## # A tibble: 30 x 5
##    Nation        Birthrate PerCapIncome PopFarms InfantMort
##    <chr>             <dbl>        <dbl>    <dbl>      <dbl>
##  1 Venezuela          46.4          392     0.4        68.5
##  2 Mexico             45.7          118     0.61       87.8
##  3 Ecuador            45.3           44     0.53      116.
##  4 Colombia           38.6          158     0.53      107.
##  5 Ceylon             37.2           81     0.53       71.6
##  6 Puerto Rico        35            374     0.37       60.2
##  7 Chile              34            187     0.3       119.
##  8 Canada             28.3          993     0.19       33.7
##  9 United States      24.7         1723     0.12       27.2
## 10 Argentina          24.7          287     0.2        62
## # i 20 more rows
```

Our goal is to find the model for birth rate using differet covariates (per capita income, proportion of population on farms, and infant mortality rate), we start with using different sets of covariates and see which correspoding model performs the best.

```r
model_1     <- as.formula("Birthrate ~ InfantMort")
model_2     <- as.formula("Birthrate ~ PopFarms")
model_3     <- as.formula("Birthrate ~ PerCapIncome")
model_4     <- as.formula("Birthrate ~ PopFarms + InfantMort")
model_5     <- as.formula("Birthrate ~ PerCapIncome + InfantMort")
model_6     <- as.formula("Birthrate ~ PerCapIncome + PopFarms")
model_full  <- as.formula("Birthrate ~ PerCapIncome + PopFarms + InfantMort")
```

```r
set.seed(123)
summary(model_1)
```

```
##  Length   Class    Mode
##       3 formula    call
```

```r
summary(model_2)
```

```
##  Length   Class    Mode
##       3 formula    call
```

```r
summary(model_3)
```

```
##  Length   Class    Mode
```

```
##       3 formula    call
```

```r
summary(model_4)
```

```
##  Length   Class    Mode
##       3 formula    call
```

```r
summary(model_5)
```

```
##  Length   Class    Mode
##       3 formula    call
```

```r
summary(model_6)
```

```
##  Length   Class    Mode
##       3 formula    call
```

```r
summary(model_full)
```

```
##  Length   Class    Mode
##       3 formula    call
```

Now, we compute 5-fold Cross Validation errors for all the models

```r
cv_mf <- cv.glm(birth_dat, glm(model_full, data = birth_dat), K = 5)$delta[1]
cv_m1 <- cv.glm(birth_dat, glm(model_1, data = birth_dat), K = 5)$delta[1]
cv_m2 <- cv.glm(birth_dat, glm(model_2, data = birth_dat), K = 5)$delta[1]
cv_m3 <- cv.glm(birth_dat, glm(model_3, data = birth_dat), K = 5)$delta[1]
cv_m4 <- cv.glm(birth_dat, glm(model_4, data = birth_dat), K = 5)$delta[1]
cv_m5 <- cv.glm(birth_dat, glm(model_5, data = birth_dat), K = 5)$delta[1]
cv_m6 <- cv.glm(birth_dat, glm(model_6, data = birth_dat), K = 5)$delta[1]
```

```r
print(paste(" Model 1: Birthrate ~ InfantMort", cv_m1))
```

```
## [1] " Model 1: Birthrate ~ InfantMort 65.199244228994"
```

```r
print(paste(" Model 2: Birthrate ~ PopFarms", cv_m2))
```

```
## [1] " Model 2: Birthrate ~ PopFarms 80.7667357024119"
```

```r
  print(paste(" Model 3: Birthrate ~ PerCapIncome", cv_m3))
```

```
## [1] " Model 3: Birthrate ~ PerCapIncome 88.0690652711383"
```

```r
print(paste(" Model 4: Birthrate ~ PopFarms + InfantMort", cv_m4))
```

```
## [1] " Model 4: Birthrate ~ PopFarms + InfantMort 62.5956219548972"
```

```r
print(paste(" Model 5: Birthrate ~ PerCapIncome + InfantMort", cv_m5))
```

```
## [1] " Model 5: Birthrate ~ PerCapIncome + InfantMort 82.5752745908939"
```

```r
print(paste(" Model 6: Birthrate ~ PerCapIncome + PopFarms", cv_m6))
```

```
## [1] " Model 6: Birthrate ~ PerCapIncome + PopFarms 84.4166801051064"
```

```r
print(paste(" Model Full: Birthrate ~ PerCapIncome + PopFarms + InfantMort", cv_mf))
```

```
## [1] " Model Full: Birthrate ~ PerCapIncome + PopFarms + InfantMort 60.688830227869"
```

We chose model 1 (Birthrate ~ InfantMort) as it has the lowest CV error of all the models.

```r
model_chosen <- glm(Birthrate ~ InfantMort, data = birth_dat)
summary(model_chosen)
```

```
##
## Call:
## glm(formula = Birthrate ~ InfantMort, data = birth_dat)
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 13.43237    2.75145   4.882 3.83e-05 ***
## InfantMort   0.21574    0.04587   4.703 6.25e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for gaussian family taken to be 53.07678)
##
##     Null deviance: 2660.1  on 29  degrees of freedom
## Residual deviance: 1486.1  on 28  degrees of freedom
## AIC: 208.22
##
## Number of Fisher Scoring iterations: 2
```

We can see here that the intercept is 13.43237 and the coefficient of InfantMort is 0.21574, So Birthrate = 13.43237 + 0.21574 * InfantMort. This shows that higher the InfantMort higher the birtrate. P-value is very small than 0.01, meaning this is significant predictor of Birthrate. I also noticed that the cv error values change when I used a different seed value for example, when seed is 567 the best model would be Birthrate ~ PopFarms + InfantMort though Birthrate ~ InfantMort has error closer to that of Birthrate ~ PopFarms + InfantMort. As errors are close, it really shouldn't make much difference in choosing between these models.

#Problem 2

```
#install.packages("pls")
library(MASS)
library(pls)
```

```
##
## Attaching package: 'pls'

## The following object is masked from 'package:stats':
##
##     loadings
```

```
data(gasoline)
summary(gasoline)
```

```
##      octane
##  Min.   :83.40
##  1st Qu.:85.88
##  Median :87.75
##  Mean   :87.18
##  3rd Qu.:88.45
##  Max.   :89.60
##        NIR.900 nm               NIR.902 nm               NIR.904 nm               NIR.906 nm
##  Min.   :-0.06283900     Min.   :-0.05623200     Min.   :-0.05307500     Min.   :-0.04815600     Min
##  1st Qu.:-0.05577350     1st Qu.:-0.05053650     1st Qu.:-0.04637875     1st Qu.:-0.04207125     1st
##  Median :-0.05354350     Median :-0.04790500     Median :-0.04393050     Median :-0.04006450     Medi
##  Mean   :-0.05282697     Mean   :-0.04743543     Mean   :-0.04358503     Mean   :-0.03918417     Mean
##  3rd Qu.:-0.05008300     3rd Qu.:-0.04427500     3rd Qu.:-0.04059125     3rd Qu.:-0.03630825     3rd
```

```
## Max.   :-0.04180600    Max.   :-0.03662100    Max.   :-0.03243000    Max.   :-0.02680700    Max
```

```r
head(gasoline)
```

```
##   octane NIR.900 nm NIR.902 nm NIR.904 nm NIR.906 nm NIR.908 nm NIR.910 nm
## 1  85.30  -0.050193  -0.045903  -0.042187  -0.037177  -0.033348  -0.031207
## 2  85.25  -0.044227  -0.039602  -0.035673  -0.030911  -0.026675  -0.023871
## 3  88.45  -0.046867  -0.041260  -0.036979  -0.031458  -0.026520  -0.023346
## 4  83.40  -0.046705  -0.042240  -0.038561  -0.034513  -0.030206  -0.027680
## 5  87.90  -0.050859  -0.045145  -0.041025  -0.036357  -0.032747  -0.031498
## 6  85.50  -0.048094  -0.042739  -0.038812  -0.034017  -0.030143  -0.027690
##   NIR.912 nm NIR.914 nm NIR.916 nm NIR.918 nm NIR.920 nm NIR.922 nm NIR.924 nm
## 1  -0.030036  -0.031298  -0.034217  -0.036012  -0.039792  -0.043037  -0.047313
## 2  -0.022571  -0.025410  -0.028960  -0.032740  -0.036683  -0.040169  -0.044899
## 3  -0.021392  -0.024993  -0.029309  -0.033920  -0.038539  -0.042571  -0.047511
## 4  -0.026042  -0.028280  -0.030920  -0.034012  -0.037082  -0.040444  -0.044858
## 5  -0.031415  -0.034611  -0.037781  -0.040752  -0.044070  -0.047320  -0.051477
## 6  -0.026387  -0.028811  -0.031481  -0.034124  -0.037468  -0.040796  -0.045207
##   NIR.926 nm NIR.928 nm NIR.930 nm NIR.932 nm NIR.934 nm NIR.936 nm NIR.938 nm
## 1  -0.048103  -0.050627  -0.053830  -0.054604  -0.056676  -0.058428  -0.060644
## 2  -0.046266  -0.048627  -0.052014  -0.053635  -0.055454  -0.056777  -0.059331
## 3  -0.048487  -0.050455  -0.053913  -0.055195  -0.056713  -0.057447  -0.059620
## 4  -0.046544  -0.048978  -0.052483  -0.054078  -0.055723  -0.056999  -0.059409
## 5  -0.052080  -0.054009  -0.057146  -0.058343  -0.059992  -0.060697  -0.062891
## 6  -0.046457  -0.048702  -0.051940  -0.053562  -0.055224  -0.056318  -0.058880
##   NIR.940 nm NIR.942 nm NIR.944 nm NIR.946 nm NIR.948 nm NIR.950 nm NIR.952 nm
## 1  -0.061712  -0.063148  -0.064480  -0.065987  -0.066894  -0.068201  -0.069434
## 2  -0.060518  -0.062084  -0.063331  -0.065059  -0.065536  -0.066877  -0.067899
## 3  -0.060656  -0.061767  -0.062859  -0.064668  -0.065566  -0.066890  -0.068397
## 4  -0.060543  -0.061868  -0.062899  -0.064401  -0.064814  -0.065989  -0.067180
## 5  -0.063776  -0.065067  -0.065707  -0.067253  -0.067910  -0.069252  -0.070794
## 6  -0.060185  -0.061395  -0.062720  -0.064315  -0.065041  -0.066414  -0.067835
##   NIR.954 nm NIR.956 nm NIR.958 nm NIR.960 nm NIR.962 nm NIR.964 nm NIR.966 nm
## 1  -0.069594  -0.071336  -0.071208  -0.071454  -0.071469  -0.071631  -0.070886
## 2  -0.067662  -0.069050  -0.068497  -0.068434  -0.067953  -0.067061  -0.066073
## 3  -0.068439  -0.070171  -0.070077  -0.070639  -0.070578  -0.070137  -0.069803
## 4  -0.066987  -0.068829  -0.069063  -0.069995  -0.070687  -0.070725  -0.070661
## 5  -0.070775  -0.072439  -0.072232  -0.072681  -0.072639  -0.072274  -0.071837
## 6  -0.067924  -0.069672  -0.069528  -0.069971  -0.070023  -0.069614  -0.069294
##   NIR.968 nm NIR.970 nm NIR.972 nm NIR.974 nm NIR.976 nm NIR.978 nm NIR.980 nm
## 1  -0.071010  -0.070609  -0.070464  -0.070433  -0.069957  -0.070208  -0.070523
## 2  -0.065564  -0.065335  -0.064051  -0.063666  -0.062513  -0.061955  -0.060906
## 3  -0.069738  -0.070031  -0.069496  -0.069789  -0.069002  -0.069288  -0.069398
## 4  -0.071043  -0.071324  -0.070894  -0.071157  -0.070798  -0.071120  -0.071266
## 5  -0.071892  -0.071883  -0.071469  -0.071713  -0.071145  -0.071399  -0.071519
## 6  -0.069277  -0.069443  -0.068892  -0.069112  -0.068510  -0.068906  -0.068990
##   NIR.982 nm NIR.984 nm NIR.986 nm NIR.988 nm NIR.990 nm NIR.992 nm NIR.994 nm
## 1  -0.070252  -0.069360  -0.068338  -0.066991  -0.065362  -0.063971  -0.062210
## 2  -0.059413  -0.058242  -0.056855  -0.055982  -0.055107  -0.054763  -0.054189
## 3  -0.068613  -0.067675  -0.066685  -0.065196  -0.063317  -0.061604  -0.059695
## 4  -0.070540  -0.069737  -0.068637  -0.067211  -0.065586  -0.064064  -0.062089
## 5  -0.070776  -0.069839  -0.068738  -0.067307  -0.065566  -0.064189  -0.062495
## 6  -0.068323  -0.067436  -0.066269  -0.065007  -0.063282  -0.061836  -0.060052
##   NIR.996 nm NIR.998 nm NIR.1000 nm NIR.1002 nm NIR.1004 nm NIR.1006 nm
## 1  -0.060678  -0.059275   -0.059126   -0.058903   -0.058488   -0.057291
```

```
## 2  -0.054361  -0.054014   -0.054822   -0.055303   -0.055101   -0.054473
## 3  -0.058403  -0.056868   -0.056419   -0.056021   -0.055192   -0.053844
## 4  -0.060639  -0.059018   -0.058185   -0.057415   -0.056311   -0.054969
## 5  -0.061220  -0.059873   -0.059576   -0.059294   -0.058375   -0.057087
## 6  -0.058887  -0.057521   -0.057181   -0.056761   -0.055913   -0.054706
##   NIR.1008 nm NIR.1010 nm NIR.1012 nm NIR.1014 nm NIR.1016 nm NIR.1018 nm
## 1   -0.055408   -0.053862   -0.051421   -0.050485   -0.048349   -0.047586
## 2   -0.053119   -0.051626   -0.048967   -0.047722   -0.045717   -0.044365
## 3   -0.051961   -0.050363   -0.047896   -0.046456   -0.044870   -0.044068
## 4   -0.052664   -0.050925   -0.048186   -0.046844   -0.045207   -0.044741
## 5   -0.055257   -0.053757   -0.051679   -0.050870   -0.049233   -0.048471
## 6   -0.052923   -0.051380   -0.049008   -0.047954   -0.046424   -0.045524
##   NIR.1020 nm NIR.1022 nm NIR.1024 nm NIR.1026 nm NIR.1028 nm NIR.1030 nm
## 1   -0.046898   -0.047726   -0.048686   -0.049626   -0.050714   -0.051541
## 2   -0.043336   -0.043581   -0.044036   -0.044856   -0.046041   -0.046858
## 3   -0.043379   -0.044227   -0.045469   -0.046493   -0.048213   -0.049097
## 4   -0.044397   -0.044697   -0.045614   -0.046452   -0.047669   -0.048569
## 5   -0.048138   -0.048762   -0.049812   -0.050631   -0.052005   -0.052855
## 6   -0.044870   -0.045398   -0.046434   -0.047223   -0.048501   -0.049288
##   NIR.1032 nm NIR.1034 nm NIR.1036 nm NIR.1038 nm NIR.1040 nm NIR.1042 nm
## 1   -0.052976   -0.054019   -0.055507   -0.056386   -0.056779   -0.056876
## 2   -0.048652   -0.049853   -0.051562   -0.052526   -0.053738   -0.054069
## 3   -0.050967   -0.052127   -0.053886   -0.054432   -0.055172   -0.055257
## 4   -0.050055   -0.051178   -0.052781   -0.053594   -0.054460   -0.054997
## 5   -0.054611   -0.055514   -0.056954   -0.057647   -0.058077   -0.058253
## 6   -0.050976   -0.052078   -0.053513   -0.054154   -0.054891   -0.054884
##   NIR.1044 nm NIR.1046 nm NIR.1048 nm NIR.1050 nm NIR.1052 nm NIR.1054 nm
## 1   -0.056439   -0.056708   -0.056432   -0.056974   -0.057091   -0.057541
## 2   -0.053925   -0.054880   -0.054857   -0.055695   -0.055656   -0.056352
## 3   -0.054866   -0.055286   -0.055256   -0.055922   -0.055649   -0.056044
## 4   -0.055201   -0.055934   -0.055994   -0.056841   -0.056687   -0.057192
## 5   -0.057819   -0.058598   -0.058334   -0.059059   -0.058660   -0.059068
## 6   -0.054520   -0.055201   -0.055025   -0.055663   -0.055174   -0.055837
##   NIR.1056 nm NIR.1058 nm NIR.1060 nm NIR.1062 nm NIR.1064 nm NIR.1066 nm
## 1   -0.057746   -0.058387   -0.058997   -0.058975   -0.059624   -0.059737
## 2   -0.056501   -0.057275   -0.057927   -0.058044   -0.058311   -0.058253
## 3   -0.056164   -0.056653   -0.057349   -0.057620   -0.058109   -0.057978
## 4   -0.057521   -0.058097   -0.058708   -0.058940   -0.059339   -0.059002
## 5   -0.058924   -0.059223   -0.059783   -0.060118   -0.060761   -0.060736
## 6   -0.055966   -0.056511   -0.057135   -0.057379   -0.057969   -0.058155
##   NIR.1068 nm NIR.1070 nm NIR.1072 nm NIR.1074 nm NIR.1076 nm NIR.1078 nm
## 1   -0.060552   -0.060416   -0.061099   -0.060784   -0.061292   -0.061811
## 2   -0.058841   -0.058138   -0.058603   -0.058309   -0.058444   -0.058993
## 3   -0.058726   -0.058442   -0.058857   -0.058779   -0.059351   -0.060286
## 4   -0.059353   -0.058568   -0.058309   -0.057284   -0.057273   -0.057433
## 5   -0.061767   -0.061507   -0.062353   -0.062139   -0.062564   -0.063118
## 6   -0.058977   -0.058727   -0.059399   -0.059298   -0.059755   -0.060350
##   NIR.1080 nm NIR.1082 nm NIR.1084 nm NIR.1086 nm NIR.1088 nm NIR.1090 nm
## 1   -0.061852   -0.062380   -0.062816   -0.063620   -0.063730   -0.064291
## 2   -0.059121   -0.059445   -0.059475   -0.060274   -0.060256   -0.060515
## 3   -0.060681   -0.061533   -0.061879   -0.062840   -0.063150   -0.063667
## 4   -0.057911   -0.059303   -0.060453   -0.062042   -0.062812   -0.063620
## 5   -0.063449   -0.064052   -0.064242   -0.065190   -0.065491   -0.065866
## 6   -0.060678   -0.061161   -0.061462   -0.062454   -0.062688   -0.063233
```

```
##   NIR.1092 nm NIR.1094 nm NIR.1096 nm NIR.1098 nm NIR.1100 nm NIR.1102 nm
## 1   -0.064209   -0.064162   -0.063681   -0.062757   -0.061833   -0.060653
## 2   -0.060839   -0.060784   -0.060401   -0.059631   -0.058887   -0.057861
## 3   -0.063744   -0.063969   -0.063588   -0.062893   -0.062346   -0.060954
## 4   -0.063369   -0.063106   -0.062210   -0.061071   -0.060476   -0.059613
## 5   -0.065748   -0.065944   -0.065408   -0.064621   -0.063747   -0.062348
## 6   -0.063210   -0.063120   -0.062835   -0.061903   -0.061317   -0.060134
##   NIR.1104 nm NIR.1106 nm NIR.1108 nm NIR.1110 nm NIR.1112 nm NIR.1114 nm
## 1   -0.059872   -0.058221   -0.057000   -0.054962   -0.052726   -0.050188
## 2   -0.057174   -0.055943   -0.054586   -0.052730   -0.050650   -0.047988
## 3   -0.060277   -0.058933   -0.057725   -0.056403   -0.054639   -0.052604
## 4   -0.059053   -0.057879   -0.056993   -0.055674   -0.053633   -0.051368
## 5   -0.061400   -0.059627   -0.058308   -0.056620   -0.054469   -0.052170
## 6   -0.059239   -0.057698   -0.056322   -0.054574   -0.052547   -0.049993
##   NIR.1116 nm NIR.1118 nm NIR.1120 nm NIR.1122 nm NIR.1124 nm NIR.1126 nm
## 1   -0.047313   -0.044383   -0.040658   -0.036981   -0.030389   -0.022454
## 2   -0.045184   -0.042775   -0.039567   -0.036483   -0.031762   -0.026061
## 3   -0.050002   -0.047952   -0.044838   -0.041595   -0.036552   -0.030222
## 4   -0.048753   -0.046565   -0.043878   -0.041106   -0.036145   -0.029643
## 5   -0.049036   -0.045807   -0.041595   -0.036657   -0.029725   -0.020507
## 6   -0.047235   -0.044684   -0.041165   -0.037443   -0.031481   -0.024195
##   NIR.1128 nm NIR.1130 nm NIR.1132 nm NIR.1134 nm NIR.1136 nm NIR.1138 nm
## 1   -0.011760    0.000941    0.018067    0.039563    0.065490    0.093330
## 2   -0.017961   -0.008352    0.004104    0.020129    0.040174    0.061190
## 3   -0.021030   -0.010311    0.003633    0.021227    0.043017    0.065780
## 4   -0.020369   -0.009442    0.005123    0.023279    0.045674    0.069009
## 5   -0.008093    0.006771    0.026072    0.050595    0.080282    0.111046
## 6   -0.013399   -0.000986    0.015126    0.035764    0.060708    0.086732
##   NIR.1140 nm NIR.1142 nm NIR.1144 nm NIR.1146 nm NIR.1148 nm NIR.1150 nm
## 1    0.119886    0.143335    0.161862    0.178692    0.190440    0.194380
## 2    0.082808    0.103309    0.121239    0.139799    0.154684    0.163895
## 3    0.088740    0.110123    0.128801    0.147827    0.163220    0.171823
## 4    0.092430    0.113654    0.131457    0.148927    0.162499    0.169830
## 5    0.140255    0.162922    0.178850    0.192453    0.200921    0.200736
## 6    0.112727    0.135831    0.154513    0.172323    0.184944    0.189945
##   NIR.1152 nm NIR.1154 nm NIR.1156 nm NIR.1158 nm NIR.1160 nm NIR.1162 nm
## 1    0.187846    0.173436    0.158487    0.146299    0.143034    0.146884
## 2    0.163702    0.155861    0.146830    0.139954    0.140520    0.147677
## 3    0.170286    0.160042    0.147558    0.137904    0.136222    0.142161
## 4    0.168042    0.159366    0.149124    0.141988    0.142149    0.148574
## 5    0.189637    0.170796    0.152340    0.138688    0.133949    0.137171
## 6    0.184606    0.171638    0.157538    0.146422    0.143336    0.147847
##   NIR.1164 nm NIR.1166 nm NIR.1168 nm NIR.1170 nm NIR.1172 nm NIR.1174 nm
## 1    0.156093    0.167857    0.179468    0.192422    0.205774    0.222009
## 2    0.159739    0.174511    0.189393    0.204829    0.222001    0.240784
## 3    0.153683    0.168846    0.184297    0.201057    0.219339    0.238474
## 4    0.159490    0.173304    0.186928    0.201306    0.217320    0.235025
## 5    0.146438    0.159196    0.172705    0.187194    0.202939    0.221091
## 6    0.157346    0.169854    0.182199    0.195600    0.211022    0.227482
##   NIR.1176 nm NIR.1178 nm NIR.1180 nm NIR.1182 nm NIR.1184 nm NIR.1186 nm
## 1    0.238851    0.257624    0.282144    0.316637    0.360689    0.410005
## 2    0.259705    0.279918    0.305844    0.342649    0.390617    0.445219
## 3    0.257398    0.277868    0.303828    0.340386    0.388889    0.444171
## 4    0.253360    0.273140    0.298527    0.334143    0.379790    0.431122
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## 5    0.239527    0.260289    0.286929    0.324194    0.370256    0.418339
## 6    0.244908    0.264326    0.288987    0.324039    0.369542    0.419514
##   NIR.1188 nm NIR.1190 nm NIR.1192 nm NIR.1194 nm NIR.1196 nm NIR.1198 nm
## 1    0.454409    0.485965    0.497682    0.489535    0.466098    0.432583
## 2    0.497316    0.535230    0.552865    0.545858    0.516133    0.474097
## 3    0.497294    0.535431    0.552290    0.541118    0.504049    0.453667
## 4    0.478623    0.512772    0.529281    0.523347    0.497873    0.462310
## 5    0.458811    0.480722    0.483581    0.465097    0.432696    0.392652
## 6    0.465683    0.498465    0.512284    0.504808    0.478875    0.442102
##   NIR.1200 nm NIR.1202 nm NIR.1204 nm NIR.1206 nm NIR.1208 nm NIR.1210 nm
## 1    0.394805    0.359732    0.331966    0.309186    0.292567    0.275393
## 2    0.429533    0.388781    0.357836    0.334257    0.316834    0.300208
## 3    0.400454    0.353163    0.318767    0.293785    0.276645    0.260872
## 4    0.423781    0.387603    0.359001    0.336412    0.319126    0.301999
## 5    0.351462    0.315084    0.288042    0.267363    0.251579    0.235251
## 6    0.402150    0.364371    0.335444    0.313017    0.295713    0.278658
##   NIR.1212 nm NIR.1214 nm NIR.1216 nm NIR.1218 nm NIR.1220 nm NIR.1222 nm
## 1    0.255452    0.235247    0.213967    0.192011    0.170424    0.150943
## 2    0.280041    0.259029    0.237029    0.215364    0.192902    0.171982
## 3    0.242538    0.224122    0.206128    0.188049    0.169174    0.152376
## 4    0.282530    0.261341    0.239522    0.218119    0.195463    0.174744
## 5    0.216596    0.198460    0.181406    0.165114    0.148187    0.133288
## 6    0.258819    0.238323    0.217383    0.196375    0.175173    0.155673
##   NIR.1224 nm NIR.1226 nm NIR.1228 nm NIR.1230 nm NIR.1232 nm NIR.1234 nm
## 1    0.132428    0.115921    0.099535    0.084481    0.071254    0.058049
## 2    0.151256    0.133448    0.115749    0.099265    0.084244    0.070886
## 3    0.136216    0.121756    0.106574    0.090796    0.076286    0.062592
## 4    0.154083    0.135921    0.118325    0.101493    0.086168    0.072287
## 5    0.118446    0.105215    0.090738    0.076775    0.063483    0.050650
## 6    0.136848    0.120534    0.104379    0.088931    0.074905    0.062107
##   NIR.1236 nm NIR.1238 nm NIR.1240 nm NIR.1242 nm NIR.1244 nm NIR.1246 nm
## 1    0.047446    0.036880    0.029082    0.021789    0.016038    0.011251
## 2    0.060095    0.049378    0.040630    0.032988    0.026985    0.021400
## 3    0.051793    0.041423    0.033518    0.026652    0.021066    0.016164
## 4    0.061218    0.050044    0.040826    0.032597    0.026030    0.020097
## 5    0.040134    0.030024    0.021975    0.015352    0.010021    0.005572
## 6    0.051506    0.040805    0.032223    0.025012    0.019011    0.014011
##   NIR.1248 nm NIR.1250 nm NIR.1252 nm NIR.1254 nm NIR.1256 nm NIR.1258 nm
## 1    0.006186    0.003116   -0.000998   -0.004703   -0.009270   -0.012907
## 2    0.016041    0.012094    0.007511    0.003694   -0.001214   -0.005412
## 3    0.011051    0.007728    0.003719   -0.000068   -0.004753   -0.008519
## 4    0.014526    0.010441    0.005578    0.001684   -0.002925   -0.007083
## 5    0.001400   -0.001737   -0.005442   -0.008756   -0.012831   -0.016278
## 6    0.009118    0.005611    0.001643   -0.001782   -0.006312   -0.010149
##   NIR.1260 nm NIR.1262 nm NIR.1264 nm NIR.1266 nm NIR.1268 nm NIR.1270 nm
## 1   -0.017234   -0.020832   -0.024051   -0.027297   -0.029792   -0.032400
## 2   -0.010057   -0.014084   -0.017764   -0.021488   -0.024172   -0.027224
## 3   -0.012552   -0.016265   -0.019766   -0.023338   -0.025883   -0.029186
## 4   -0.011555   -0.015370   -0.018619   -0.021719   -0.023654   -0.025991
## 5   -0.020192   -0.023579   -0.027011   -0.030106   -0.032377   -0.035369
## 6   -0.014581   -0.018146   -0.021717   -0.024962   -0.027266   -0.030051
##   NIR.1272 nm NIR.1274 nm NIR.1276 nm NIR.1278 nm NIR.1280 nm NIR.1282 nm
## 1   -0.033746   -0.035721   -0.036701   -0.037363   -0.037377   -0.037489
## 2   -0.029156   -0.031021   -0.032030   -0.032565   -0.032049   -0.031394
```

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## 3   -0.031037   -0.033466   -0.034469   -0.035675   -0.035766   -0.036086
## 4   -0.027169   -0.028784   -0.029554   -0.030701   -0.031122   -0.031775
## 5   -0.036933   -0.038394   -0.039402   -0.040477   -0.040559   -0.040550
## 6   -0.031876   -0.033743   -0.034756   -0.035539   -0.035698   -0.035932
##   NIR.1284 nm NIR.1286 nm NIR.1288 nm NIR.1290 nm NIR.1292 nm NIR.1294 nm
## 1   -0.037695   -0.036906   -0.037094   -0.036752   -0.037211   -0.037308
## 2   -0.031097   -0.029836   -0.029678   -0.028968   -0.029269   -0.029262
## 3   -0.036513   -0.036003   -0.036171   -0.035911   -0.036362   -0.036318
## 4   -0.032641   -0.032724   -0.033748   -0.034132   -0.035013   -0.035620
## 5   -0.040637   -0.040203   -0.040332   -0.039843   -0.039822   -0.039774
## 6   -0.036067   -0.035340   -0.035486   -0.035167   -0.035553   -0.035704
##   NIR.1296 nm NIR.1298 nm NIR.1300 nm NIR.1302 nm NIR.1304 nm NIR.1306 nm
## 1   -0.037673   -0.037877   -0.038212   -0.038426   -0.038534   -0.039103
## 2   -0.029877   -0.030030   -0.030565   -0.031404   -0.032320   -0.033693
## 3   -0.036997   -0.037092   -0.037218   -0.037411   -0.037429   -0.038010
## 4   -0.036407   -0.036595   -0.036742   -0.037076   -0.037230   -0.037877
## 5   -0.040008   -0.040040   -0.040031   -0.040196   -0.040369   -0.041101
## 6   -0.036136   -0.036285   -0.036488   -0.036742   -0.036819   -0.037475
##   NIR.1308 nm NIR.1310 nm NIR.1312 nm NIR.1314 nm NIR.1316 nm NIR.1318 nm
## 1   -0.038933   -0.039611   -0.039168   -0.038598   -0.037733   -0.037027
## 2   -0.034155   -0.035496   -0.035629   -0.035932   -0.035289   -0.034887
## 3   -0.037523   -0.038127   -0.037544   -0.037255   -0.036148   -0.035291
## 4   -0.037530   -0.038428   -0.037948   -0.037560   -0.036566   -0.035762
## 5   -0.040674   -0.041522   -0.040715   -0.040219   -0.039039   -0.037941
## 6   -0.037292   -0.037909   -0.037347   -0.037000   -0.035950   -0.035043
##   NIR.1320 nm NIR.1322 nm NIR.1324 nm NIR.1326 nm NIR.1328 nm NIR.1330 nm
## 1   -0.035891   -0.034290   -0.033224   -0.031475   -0.030071   -0.028086
## 2   -0.034467   -0.033427   -0.032634   -0.030815   -0.029493   -0.027298
## 3   -0.034674   -0.033358   -0.032758   -0.031228   -0.030097   -0.027881
## 4   -0.035156   -0.033697   -0.032653   -0.030570   -0.028914   -0.026203
## 5   -0.037409   -0.036171   -0.035326   -0.033454   -0.032407   -0.030240
## 6   -0.034492   -0.033096   -0.032354   -0.030384   -0.029077   -0.026629
##   NIR.1332 nm NIR.1334 nm NIR.1336 nm NIR.1338 nm NIR.1340 nm NIR.1342 nm
## 1   -0.026779   -0.024450   -0.022709   -0.020088   -0.016814   -0.013638
## 2   -0.026081   -0.024199   -0.022546   -0.020154   -0.017032   -0.013717
## 3   -0.026697   -0.024809   -0.022950   -0.020248   -0.016845   -0.013423
## 4   -0.024685   -0.022470   -0.020930   -0.018910   -0.016121   -0.013279
## 5   -0.029053   -0.026720   -0.024568   -0.021595   -0.017934   -0.013938
## 6   -0.025188   -0.023047   -0.021273   -0.018913   -0.015700   -0.012374
##   NIR.1344 nm NIR.1346 nm NIR.1348 nm NIR.1350 nm NIR.1352 nm NIR.1354 nm
## 1   -0.008755   -0.003402    0.004585    0.015708    0.032342    0.055075
## 2   -0.008264   -0.002064    0.007304    0.020121    0.039083    0.064193
## 3   -0.007559   -0.000991    0.008786    0.022739    0.043209    0.069797
## 4   -0.008334   -0.003372    0.004753    0.016161    0.033429    0.056411
## 5   -0.008015   -0.001863    0.007269    0.019779    0.038201    0.062477
## 6   -0.007267   -0.001764    0.006434    0.017937    0.035312    0.058297
##   NIR.1356 nm NIR.1358 nm NIR.1360 nm NIR.1362 nm NIR.1364 nm NIR.1366 nm
## 1    0.083261    0.114275    0.142234    0.165117    0.181014    0.194140
## 2    0.095086    0.128027    0.158800    0.183265    0.201249    0.216701
## 3    0.102595    0.137954    0.169100    0.194530    0.212629    0.227677
## 4    0.085082    0.116397    0.145471    0.168918    0.185935    0.200819
## 5    0.092206    0.123556    0.151961    0.174343    0.189944    0.202222
## 6    0.087569    0.119281    0.148086    0.171700    0.188415    0.202311
##   NIR.1368 nm NIR.1370 nm NIR.1372 nm NIR.1374 nm NIR.1376 nm NIR.1378 nm
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## 1    0.207728    0.225084    0.245205    0.267331    0.290558    0.311237
## 2    0.232922    0.252380    0.274944    0.298853    0.322715    0.342958
## 3    0.243204    0.262941    0.285764    0.309733    0.333065    0.352530
## 4    0.215868    0.234213    0.255771    0.279190    0.303732    0.325323
## 5    0.215936    0.233155    0.254445    0.277749    0.301495    0.320695
## 6    0.216533    0.233879    0.254700    0.277301    0.300612    0.320822
##   NIR.1380 nm NIR.1382 nm NIR.1384 nm NIR.1386 nm NIR.1388 nm NIR.1390 nm
## 1    0.328961    0.341901    0.350544    0.357751    0.364172    0.370471
## 2    0.360426    0.372804    0.381960    0.389191    0.397112    0.403062
## 3    0.369372    0.380216    0.387849    0.394677    0.402078    0.408582
## 4    0.344871    0.359555    0.370893    0.378461    0.386247    0.392268
## 5    0.336187    0.345160    0.350677    0.355522    0.362320    0.368313
## 6    0.338919    0.351633    0.360320    0.367231    0.374814    0.380750
##   NIR.1392 nm NIR.1394 nm NIR.1396 nm NIR.1398 nm NIR.1400 nm NIR.1402 nm
## 1    0.373154    0.368176    0.356007    0.341120    0.325587    0.313571
## 2    0.406006    0.401295    0.389064    0.373591    0.356204    0.342050
## 3    0.410932    0.406170    0.393839    0.376215    0.357096    0.340687
## 4    0.393815    0.388471    0.376881    0.360814    0.344653    0.331744
## 5    0.370052    0.365085    0.352979    0.337215    0.321339    0.308459
## 6    0.383805    0.378930    0.366319    0.351337    0.335220    0.322073
##   NIR.1404 nm NIR.1406 nm NIR.1408 nm NIR.1410 nm NIR.1412 nm NIR.1414 nm
## 1    0.306847    0.305310    0.304471    0.304125    0.298698    0.285589
## 2    0.332780    0.329005    0.328951    0.328051    0.322452    0.308979
## 3    0.329190    0.322999    0.321164    0.319110    0.311924    0.298671
## 4    0.323812    0.321500    0.323119    0.322928    0.317972    0.305534
## 5    0.299508    0.295653    0.295204    0.292860    0.286862    0.274217
## 6    0.314083    0.311147    0.312599    0.311080    0.304679    0.291074
##   NIR.1416 nm NIR.1418 nm NIR.1420 nm NIR.1422 nm NIR.1424 nm NIR.1426 nm
## 1    0.267933    0.247573    0.229774    0.215690    0.207192    0.202051
## 2    0.290704    0.269241    0.249374    0.233860    0.224821    0.220504
## 3    0.280373    0.258819    0.238800    0.222677    0.213798    0.209824
## 4    0.288311    0.267978    0.248524    0.232934    0.223277    0.217777
## 5    0.257335    0.237130    0.219594    0.205760    0.197859    0.195243
## 6    0.274145    0.253419    0.234735    0.219703    0.211142    0.206869
##   NIR.1428 nm NIR.1430 nm NIR.1432 nm NIR.1434 nm NIR.1436 nm NIR.1438 nm
## 1    0.201819    0.202353    0.201612    0.199291    0.193770    0.186669
## 2    0.220685    0.222163    0.222347    0.221434    0.215752    0.209041
## 3    0.210649    0.212908    0.214331    0.214048    0.209266    0.202558
## 4    0.216610    0.216723    0.216083    0.214255    0.208305    0.201097
## 5    0.196015    0.197585    0.198133    0.196885    0.190623    0.184152
## 6    0.206575    0.207201    0.206917    0.205378    0.199688    0.192726
##   NIR.1440 nm NIR.1442 nm NIR.1444 nm NIR.1446 nm NIR.1448 nm NIR.1450 nm
## 1    0.178426    0.171231    0.164144    0.157259    0.149194    0.139789
## 2    0.200297    0.192172    0.183974    0.175816    0.167254    0.156358
## 3    0.193675    0.185359    0.176963    0.169181    0.159964    0.149142
## 4    0.192070    0.183795    0.176042    0.168643    0.160181    0.150664
## 5    0.175524    0.167989    0.159801    0.152625    0.144600    0.134160
## 6    0.184129    0.176453    0.169165    0.161874    0.153787    0.143930
##   NIR.1452 nm NIR.1454 nm NIR.1456 nm NIR.1458 nm NIR.1460 nm NIR.1462 nm
## 1    0.130173    0.120745    0.113733    0.107519    0.103765    0.101175
## 2    0.146486    0.137275    0.130180    0.123365    0.119438    0.116286
## 3    0.139082    0.129739    0.122022    0.116297    0.113173    0.110954
## 4    0.141494    0.133234    0.126515    0.120700    0.117645    0.115052
## 5    0.125008    0.115873    0.109185    0.103171    0.100027    0.098121
```

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## 6    0.134341    0.124761    0.117305    0.111363    0.107833    0.105109
##    NIR.1464 nm NIR.1466 nm NIR.1468 nm NIR.1470 nm NIR.1472 nm NIR.1474 nm
## 1    0.100106    0.099068    0.097315    0.095724    0.093093    0.090002
## 2    0.114482    0.112444    0.108714    0.106003    0.102389    0.098235
## 3    0.110095    0.109058    0.106665    0.104405    0.100941    0.097474
## 4    0.113629    0.112397    0.108961    0.106436    0.102798    0.098874
## 5    0.097745    0.097652    0.096193    0.095738    0.093557    0.090709
## 6    0.103951    0.103030    0.100363    0.098874    0.095930    0.092720
##    NIR.1476 nm NIR.1478 nm NIR.1480 nm NIR.1482 nm NIR.1484 nm NIR.1486 nm
## 1    0.086612    0.084922    0.080780    0.077849    0.074536    0.070910
## 2    0.093959    0.091774    0.087636    0.084646    0.082097    0.079089
## 3    0.093497    0.091673    0.087493    0.084593    0.081474    0.078340
## 4    0.094944    0.092621    0.088422    0.084868    0.081632    0.078056
## 5    0.087039    0.085000    0.080375    0.077113    0.073986    0.069980
## 6    0.089492    0.088125    0.084145    0.080995    0.078229    0.074285
##    NIR.1488 nm NIR.1490 nm NIR.1492 nm NIR.1494 nm NIR.1496 nm NIR.1498 nm
## 1    0.068700    0.064584    0.061726    0.057395    0.053704    0.049329
## 2    0.077735    0.074444    0.071813    0.068586    0.065580    0.061501
## 3    0.076682    0.073916    0.071351    0.068287    0.065568    0.061216
## 4    0.076356    0.072892    0.069921    0.066060    0.062555    0.057895
## 5    0.067515    0.064092    0.060768    0.057105    0.054163    0.050508
## 6    0.071997    0.068224    0.065002    0.061145    0.058086    0.053640
##    NIR.1500 nm NIR.1502 nm NIR.1504 nm NIR.1506 nm NIR.1508 nm NIR.1510 nm
## 1    0.044695    0.039937    0.034641    0.030164    0.024898    0.021588
## 2    0.056740    0.051068    0.045104    0.040192    0.034145    0.029970
## 3    0.055821    0.049529    0.043150    0.037184    0.030828    0.026308
## 4    0.052995    0.047908    0.042637    0.038187    0.032880    0.029345
## 5    0.045830    0.040213    0.034258    0.029202    0.023555    0.019996
## 6    0.048868    0.043618    0.038088    0.033546    0.028124    0.024610
##    NIR.1512 nm NIR.1514 nm NIR.1516 nm NIR.1518 nm NIR.1520 nm NIR.1522 nm
## 1    0.017102    0.014860    0.010854    0.008205    0.005869    0.003920
## 2    0.024131    0.020670    0.016931    0.013083    0.010199    0.007926
## 3    0.020805    0.016907    0.013194    0.009759    0.007271    0.004796
## 4    0.024026    0.020904    0.017431    0.014136    0.011749    0.009634
## 5    0.015186    0.012201    0.008836    0.005561    0.003463    0.001051
## 6    0.019739    0.016248    0.012841    0.009775    0.007571    0.005427
##    NIR.1524 nm NIR.1526 nm NIR.1528 nm NIR.1530 nm NIR.1532 nm NIR.1534 nm
## 1    0.002107   -0.000203   -0.001072   -0.003325   -0.004707   -0.006635
## 2    0.005733    0.002833    0.001596   -0.000928   -0.002545   -0.003656
## 3    0.002720    0.000243   -0.001196   -0.003965   -0.005704   -0.007156
## 4    0.007699    0.005545    0.004693    0.002553    0.001494   -0.000102
## 5   -0.001164   -0.003490   -0.004459   -0.007406   -0.009019   -0.010111
## 6    0.003552    0.001367    0.000394   -0.002218   -0.003748   -0.004834
##    NIR.1536 nm NIR.1538 nm NIR.1540 nm NIR.1542 nm NIR.1544 nm NIR.1546 nm
## 1   -0.008403   -0.009317   -0.011351   -0.012280   -0.013474   -0.014453
## 2   -0.006217   -0.006885   -0.008307   -0.009142   -0.010409   -0.010983
## 3   -0.009860   -0.010959   -0.012194   -0.013605   -0.014564   -0.015313
## 4   -0.003056   -0.004391   -0.006868   -0.008665   -0.010355   -0.011261
## 5   -0.012779   -0.013826   -0.015100   -0.015879   -0.017178   -0.017586
## 6   -0.007641   -0.008543   -0.010223   -0.011276   -0.012683   -0.013216
##    NIR.1548 nm NIR.1550 nm NIR.1552 nm NIR.1554 nm NIR.1556 nm NIR.1558 nm
## 1   -0.015262   -0.015588   -0.016349   -0.016028   -0.016209   -0.015985
## 2   -0.012033   -0.012020   -0.013442   -0.012709   -0.011563   -0.010432
## 3   -0.015723   -0.015263   -0.016537   -0.015581   -0.014147   -0.013280
```

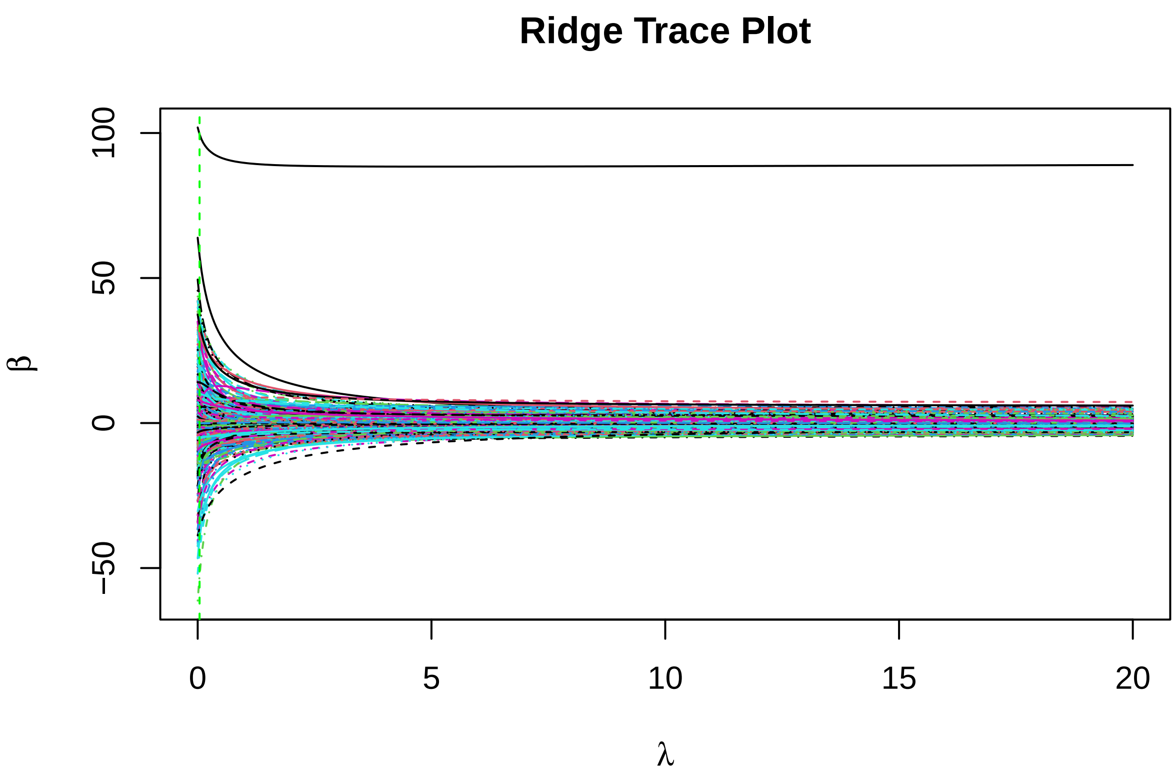
```
## 4   -0.011896   -0.011357   -0.012562   -0.011408   -0.010164   -0.009194
## 5   -0.018151   -0.017935   -0.019162   -0.018361   -0.017309   -0.016847
## 6   -0.014205   -0.013976   -0.015312   -0.014756   -0.013938   -0.012932
##   NIR.1560 nm NIR.1562 nm NIR.1564 nm NIR.1566 nm NIR.1568 nm NIR.1570 nm
## 1   -0.015483   -0.014216   -0.013790   -0.013844   -0.011773   -0.010432
## 2   -0.010091   -0.007960   -0.006697   -0.005367   -0.003432   -0.002665
## 3   -0.013291   -0.012268   -0.011176   -0.010897   -0.009383   -0.008983
## 4   -0.009642   -0.008700   -0.007899   -0.006986   -0.005475   -0.004838
## 5   -0.016776   -0.015592   -0.014585   -0.014661   -0.013113   -0.013092
## 6   -0.012850   -0.011878   -0.011203   -0.010540   -0.009494   -0.009448
##   NIR.1572 nm NIR.1574 nm NIR.1576 nm NIR.1578 nm NIR.1580 nm NIR.1582 nm
## 1   -0.009606   -0.008209   -0.006732   -0.004573   -0.003433   -0.001200
## 2   -0.001307   -0.000406    0.001300    0.003107    0.003075    0.004513
## 3   -0.007542   -0.005716   -0.003888   -0.001115    0.000176    0.002464
## 4   -0.003785   -0.002646   -0.001373    0.000559    0.001422    0.003357
## 5   -0.011257   -0.010118   -0.007846   -0.005406   -0.004259   -0.002039
## 6   -0.007761   -0.006355   -0.004604   -0.002245   -0.001394    0.000746
##   NIR.1584 nm NIR.1586 nm NIR.1588 nm NIR.1590 nm NIR.1592 nm NIR.1594 nm
## 1   -0.000066    0.002759    0.004036    0.006620    0.008709    0.011810
## 2    0.005336    0.008496    0.009262    0.010982    0.012330    0.015301
## 3    0.003951    0.008541    0.009676    0.012153    0.013639    0.016898
## 4    0.005481    0.010048    0.011528    0.014357    0.015642    0.018304
## 5   -0.000655    0.003930    0.004579    0.007207    0.008612    0.012129
## 6    0.002009    0.005896    0.006818    0.009577    0.010808    0.014127
##   NIR.1596 nm NIR.1598 nm NIR.1600 nm NIR.1602 nm NIR.1604 nm NIR.1606 nm
## 1    0.014426    0.017254    0.020633    0.025032    0.029496    0.034242
## 2    0.018270    0.020912    0.024617    0.028188    0.033321    0.037755
## 3    0.019847    0.022470    0.026069    0.029714    0.034281    0.039133
## 4    0.020612    0.022399    0.025038    0.027860    0.032481    0.036772
## 5    0.015541    0.018535    0.022522    0.026739    0.032223    0.037103
## 6    0.017126    0.019856    0.023354    0.027172    0.032375    0.036398
##   NIR.1608 nm NIR.1610 nm NIR.1612 nm NIR.1614 nm NIR.1616 nm NIR.1618 nm
## 1    0.039735    0.046803    0.053743    0.062788    0.073531    0.087246
## 2    0.043515    0.050530    0.058033    0.067009    0.077781    0.091191
## 3    0.045228    0.051833    0.059036    0.068655    0.079319    0.092339
## 4    0.042282    0.049100    0.056030    0.065351    0.075811    0.089083
## 5    0.043411    0.051336    0.059517    0.069594    0.081145    0.096240
## 6    0.042162    0.048944    0.056253    0.065159    0.075938    0.089891
##   NIR.1620 nm NIR.1622 nm NIR.1624 nm NIR.1626 nm NIR.1628 nm NIR.1630 nm
## 1    0.104000    0.124210    0.147924    0.175024    0.202711    0.232547
## 2    0.106815    0.123946    0.146013    0.171884    0.197074    0.224837
## 3    0.107952    0.123435    0.143631    0.166498    0.188171    0.210848
## 4    0.105323    0.123006    0.145753    0.173142    0.200956    0.231804
## 5    0.112862    0.131164    0.153401    0.178326    0.202094    0.228229
## 6    0.106788    0.124578    0.148279    0.174457    0.201765    0.231973
##   NIR.1632 nm NIR.1634 nm NIR.1636 nm NIR.1638 nm NIR.1640 nm NIR.1642 nm
## 1    0.259828    0.282117    0.299937    0.313416    0.322870    0.327217
## 2    0.250088    0.270191    0.284011    0.290728    0.294182    0.293247
## 3    0.232041    0.249246    0.261166    0.271317    0.276957    0.279662
## 4    0.259687    0.281929    0.297769    0.308358    0.311430    0.311149
## 5    0.252492    0.273077    0.290969    0.307760    0.321736    0.331647
## 6    0.258594    0.281735    0.299311    0.311416    0.319700    0.322975
##   NIR.1644 nm NIR.1646 nm NIR.1648 nm NIR.1650 nm NIR.1652 nm NIR.1654 nm
## 1    0.331907    0.336522    0.341847    0.350585    0.368032    0.393494
```

```
## 2     0.289916    0.291554    0.293929    0.301816    0.316670    0.336533
## 3     0.280366    0.284626    0.289691    0.297987    0.312909    0.333932
## 4     0.308663    0.308939    0.311825    0.319857    0.335700    0.355742
## 5     0.337756    0.344823    0.352189    0.362563    0.379202    0.404428
## 6     0.324616    0.327259    0.332877    0.343881    0.360419    0.383758
##   NIR.1656 nm NIR.1658 nm NIR.1660 nm NIR.1662 nm NIR.1664 nm NIR.1666 nm
## 1     0.427370    0.472357    0.528721    0.593168    0.666647    0.743945
## 2     0.363209    0.399271    0.443964    0.499024    0.559683    0.624577
## 3     0.362284    0.400352    0.445686    0.504338    0.567039    0.635874
## 4     0.384754    0.424769    0.472003    0.531050    0.596606    0.662997
## 5     0.441534    0.494011    0.557655    0.635493    0.721107    0.803044
## 6     0.417324    0.460072    0.513475    0.577686    0.649837    0.724448
##   NIR.1668 nm NIR.1670 nm NIR.1672 nm NIR.1674 nm NIR.1676 nm NIR.1678 nm
## 1     0.833972    0.909368    0.985332    1.051159    1.107395    1.158488
## 2     0.697450    0.770433    0.848769    0.916590    0.975048    1.036499
## 3     0.711690    0.792939    0.872021    0.938915    1.004690    1.067934
## 4     0.744126    0.817398    0.895422    0.959761    1.021595    1.082497
## 5     0.893778    0.968985    1.050312    1.104573    1.152964    1.192131
## 6     0.807390    0.888981    0.964555    1.038897    1.098220    1.142596
##   NIR.1680 nm NIR.1682 nm NIR.1684 nm NIR.1686 nm NIR.1688 nm NIR.1690 nm
## 1     1.174795    1.198461    1.224243    1.242645    1.250789    1.246626
## 2     1.085067    1.128877    1.148342    1.189116    1.223242    1.253306
## 3     1.108813    1.147964    1.167798    1.198287    1.237383    1.260979
## 4     1.117447    1.160089    1.169350    1.201066    1.233299    1.262966
## 5     1.219254    1.252712    1.238013    1.259616    1.273713    1.296524
## 6     1.179569    1.214046    1.210217    1.241090    1.262138    1.288401
##   NIR.1692 nm NIR.1694 nm NIR.1696 nm NIR.1698 nm NIR.1700 nm
## 1     1.250985    1.264189    1.244678    1.245913    1.221135
## 2     1.282889    1.215065    1.225211    1.227985    1.198851
## 3     1.276677    1.218871    1.223132    1.230321    1.208742
## 4     1.272709    1.211068    1.215044    1.232655    1.206696
## 5     1.299507    1.226448    1.230718    1.232864    1.202926
## 6     1.291118    1.229769    1.227615    1.227630    1.207576
```

```r
regression_model<- lm.ridge(octane ~ ., data = gasoline, lambda = seq(0, 20, length = 500))
best_lambda_index <- which.min(regression_model$GCV)
best_lambda <- regression_model$lambda[best_lambda_index]
```

#Part a(1)

```r
ridge_coefficients <- t(coef(regression_model))
matplot(regression_model$lambda, coef(regression_model), type = "l",
        xlab = expression(lambda), ylab = expression(hat(beta)),
        main = "Ridge Trace Plot")
abline(v = best_lambda, col = "green", lty = 2)
```



#Part a(2)

```
best_ridge_coefficients <- coef(regression_model)[, best_lambda_index]
nonzero_est_coeff <- sum(best_ridge_coefficients != 0)
print(paste("Number of Non Zero estimated coefficients is ", nonzero_est_coeff ))
```

```
## [1] "Number of Non Zero estimated coefficients is  500"
```

#Part a(3)

```
print(as.vector(best_ridge_coefficients[1:5]))
```

```
## [1] -13.861456 -11.684332 -10.150145  -9.049931  -8.238839
```

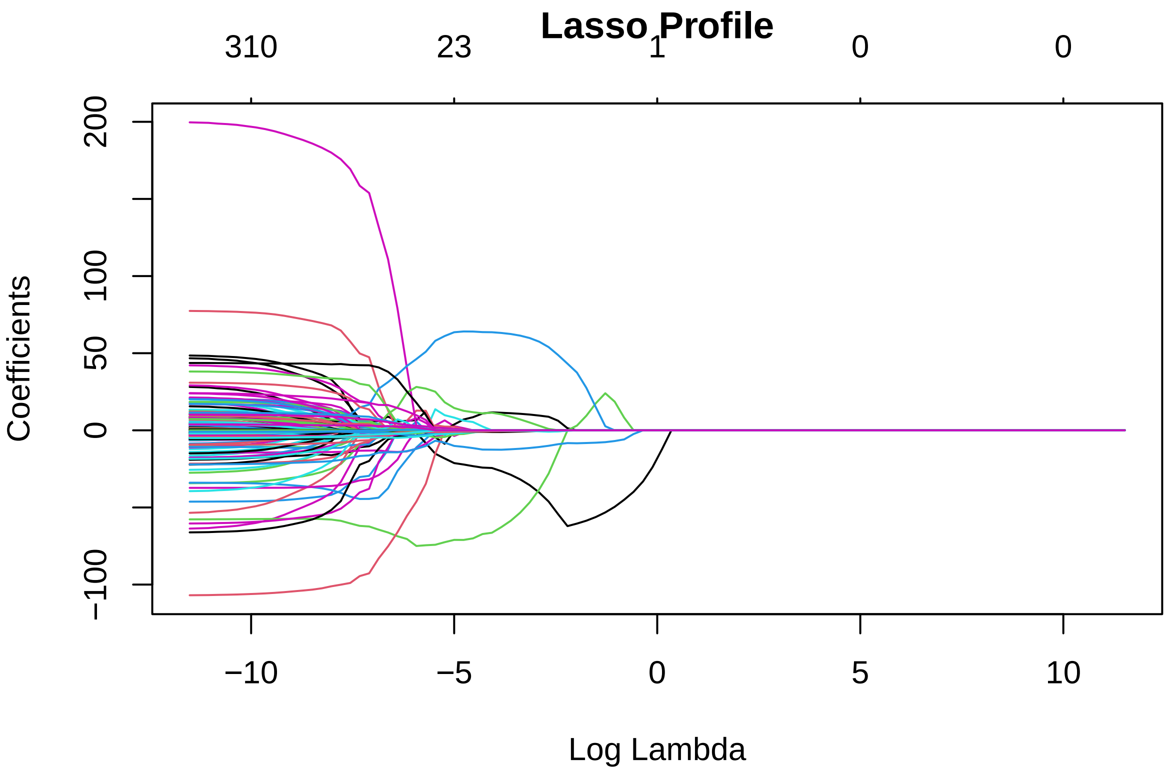
#Part b (1)

```
#install.packages('glmnet')
library(glmnet)
```

```
## Loading required package: Matrix

## Loaded glmnet 4.1-8
```

```
#Based on lecture set 04 slide 51
Xfull <- model.matrix(octane ~ ., data = gasoline)
Xfull <- Xfull[, -1]
Y <- gasoline$octane
lambdas <- 10^{seq(from = -5, to = 5, length = 100)}
lafit <- glmnet(Xfull, Y, alpha = 1, lambda = lambdas)
plot(lafit, xvar = "lambda", main = "Lasso Profile")
```



#Part b(2)

```
las_cv <- cv.glmnet(Xfull, Y, alpha = 1)
nocoef <- coef(las_cv, s = "lambda.min")
nonzero_coef <- sum(nocoef[-1] != 0)
print(paste("Number of Non Zero coefficients is ", nonzero_coef ))
```

```
## [1] "Number of Non Zero coefficients is  14"
```

#Part b(3)

```
print(as.vector(nocoef[1:5]))
```

```
## [1] 96.94275  0.00000  0.00000  0.00000  0.00000
```

#Part c As we can see from the coefficients, ridge regression keeps all the variables where as lasso has some coefficients zero meaning ridge would be better when all covariate correspod to the response, Lasso is better when only few variables are important.

#Problem 3 # Part 1 (Problem 3 from Chapter 6 of Introduction to Statistical Learning with Applications in R) #part a (iv) As we increase s from zero, the training RSS will steadily decrease. As s increases, more coefficients become non zero, this makes model fit better thus steadily decreasing RSS.

#part b (ii) The test RSS will Decrease initially, and then start increasing in U shape as we increase s from zero. As mentioned in part a, initially it decreases because the model fits better but after a point of s, the model will overfit leading to increase in RSS.

#part c (iii) As s increases, more coefficients become non zero, leading to increase in variace steadily.

#part d (iv) As we increase s from zero, the model will go from underfitting to over fitting, this makes the bias steadily decrease.

#part e (v) Irreducivle error remains constant, as it is the noise in data and isn't effected by s.

#Part 2(Problem 4 from Chapter 6 of Introduction to Statistical Learning with Applications in R) #part a

(iii) Here, it is ridge. as we increase lambda the model becomes less flexible and can't fit properly to the data leading to steady increase in RSS.

#part b (ii) small lambda as mentioned earlier leads to high RSS, as lambda increases the test RSS will decrease till we reach optimal lambda and once we cross it the test RSS will steadily increase again due to under fitting giving us a U - shape.

#part c (iv) variace decrreases steadily, as we increase lambda model becomes simpler and at high values it under fits, this leads to gradual decrease in variance.

#part d (iii) as mentioned before, increasing to high lambda leads to under fitting, as we move towards a simpler model making the bias steadily increase.

#part e (v) just like with lasso, ridge also has irreducible error constant as it is due to noise and is not effected by change in lambda value.

PROBLEM 4

(a) We want to find probability a randomly selected student from the US smokes ($P(Y=1)$)

$$\hat{p} = \frac{\text{Total students that smoke}}{\text{Total no. of students}}$$

$$= \frac{90+21}{1000} = \frac{111}{1000} = 0.111$$

(b) We want to calculate an estimate for probability a randomly selected student from the US whose parents/ guardian smokes also smokes ($P(Y=1|X=1)$)

$$\hat{p} = \frac{\text{no of students smoke \& atleast 1 parent/ guardian smokes}}{\text{no. of students whose parents smoke}}$$

$$= \frac{90}{90+238} = \frac{90}{328} = 0.274$$

We want probability that a randomly selected student from the US smokes given their parents/ guardian donot smoke
($P(Y=1|X=0)$)

$P(Y=1|x=0)$ = $\dfrac{\text{no. of students smoke but parents/guardians donot smoke}}{\text{no. of students whose parents donot smoke}}$

$= \dfrac{21}{21+651} = \dfrac{21}{672} = 0.031$

(c) odds of $Y=1$ when $x=1$ $= \dfrac{P(Y=1|x=1)}{1-P(Y=1|x=1)}$

$= \dfrac{0.274}{1-0.274}$

$= \dfrac{0.274}{0.726} = 0.377$

(d) For every one student with atleast one smoking parent who doesn't smoke, 0.377 students do smoke.

e odds of $Y=1$ when $x=0$, $= \dfrac{P(Y=1|x=0)}{1-P(Y=1|x=0)}$

$= \dfrac{0.031}{0.969} = 0.032$

(f) for students whose parents donot smoke, for every 1 student that doesn't smoke 0.032 students smoke.

9) As discussed in class,
The log' model introduced for $Y_i \sim Bern(\pi(X_i))$
with Y_i's being independent is

$$\log\left(\frac{\pi(x_i)}{1-\pi(x_i)}\right) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$$

where $E(Y_i|x_i) = \pi(X_i)$

log odds of student smoking based on whether they have parent/guardian who smokes.

This is ~~X=1~~, ~~X=1~~

$Y = 0/1$ (we need to predict this)

based on X.

$$\log\left(\frac{\pi(x_i)}{1-\pi(x_i)}\right) = \beta_0 + \beta_1 x$$ (we have only one covariate)

$$\log\left(\frac{P(Y=1|x)}{1-P(Y=1|x)}\right) = \beta_0 + \beta_1 x$$

β_1 is the log of ratio of odds for $Y=1$ when $x=1$ to odds of $Y=1$ when $x=0$.

$$\beta_1 = \log\left(\frac{\text{odds when } Y=1, x=1}{\text{odds when } Y=1, x=0}\right) = \log\left(\frac{0.377}{0.032}\right)$$

(from part c fe

est' $\beta_1 = \log(11.78)$

This means for 1 unit increase in x, there will be $\log(11.78)$ units increase in log of odds of student smoking $\left(\frac{\text{odds when } Y=1, x=1}{\text{odds when } Y=1, x=0} \right)$